



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VIII

Subject Name: Biostatistics and Research Methodology

Subject Code: BP801T

Max. Marks: 75

Date: 12/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- i) The coefficient of correlation is independent of _____
- a) change of scale
 - b) change of origin
 - c) change of scale and origin
 - d) neither change of scale nor origin
- ii) While calculating the standard deviation, the deviations are only taken from the _____
- a) mode value of a series
 - b) median value of a series
 - c) quartile value of a series
 - d) mean value of a series
- iii) The numerical value of a standard deviation can never be _____.
- a) negative
 - b) Zero
 - c) larger than the variance
 - d) larger than the deviation
- iv) One of the followings measures of central tendency includes the magnitude of scores
- a) Mean
 - b) Mode
 - c) Median
 - d) Range
- v) In the least square method, we use _____ to find the value of unknowns.
- a) Regression equations
 - b) Normal equations
 - c) General equations
 - d) Auxiliary equations
- vi) In regression, the equation that describes how the response variable (y) is related to the explanatory variable (x) is the _____
- a) correlation model
 - b) regression model
 - c) correlation coefficient
 - d) factorial model



vii) In regression analysis, the variable that is being predicted is the

- a) response, or dependent, variable
- b) independent variable
- c) Intervening variable
- d) is usually x

viii) In a Binomial Distribution, if 'n' is the number of trials and 'p' is the probability of success, then the mean value is given by _____

- a) np
- b) n
- c) p
- d) np(1-p)

ix) The Wilcoxon rank-sum test is a

- a) parametric
- b) nonparametric
- c) regression
- d) Correlation

x) A popular nonparametric test to compare outcomes between two independent groups is

- a) Wilcoxon Rank Sum Test
- b) Kruskal-Wallis
- c) Mann Whitney U test
- d) ANOVA

xi) Wilcoxon signed-rank test can be used as an alternative to

- a) the paired Student's t-test
- b) the paired Student's p-test
- c) Chi sq test
- d) t or p test

xii) Research is

- a) Repeated Search
- b) Finding solution to problem
- c) Working in a scientific way to search for truth of any problem
- d) Doing literature review

xiii) Action research means

- a) longitudinal research
- b) an analytical research
- c) a research initiated to solve an immediate problem
- d) non analytical research

xiv) Last stage of research process is

- a) Literature review
- b) Report preparation
- c) Publication
- d) Presentation

xv) Blocking is the arranging of experimental units in _____ that are similar to one another.

- a) pairing
- b) groups
- c) blocks
- d) groups and blocks

xvi) Blocking reduces unexplained

a) similarity

b) variability

c) problems

d) errors

xvii) Two level factorial experiments are factorial experiments in which each factor is investigated at _____ levels

a) single

b) two

c) multiple

d) three

xviii) The term Optimization is defined as to make experiment

a) error free

b) reliable

c) perfect, effective, and functional as possible

d) authentic

xix) Fractional factorial designs are used

a) for small set of factors

b) to examine multiple factors efficiently with fewer runs than corresponding full factorial design

c) for single factor efficiently with fewer runs

d) to examine variable

xx) In the Central Composite design arrays, the levels of the factors are on the

a) center and circumscribed at the center of side

b) edges, center and circumscribed at the center of side

c) edges and center of side

d) edges only

Q. 2. Long Answers) = $2 \times 10 = 20$ (Answer 2 out of 3)

i) Explain QbD. DoE is essential in application of QbD in product development process, explain.

ii) Elaborate different phases of clinical trial.

iii) Explain the Measures of Dispersion.

Calculate Mean, Variance and Standard Deviation for the given data.

Height (cm)	135-140	141-150	146-150	151-155	156-160	161-165	166-170
Frequency	08	12	18	22	20	14	10

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

- i) Explain ANOVA and its applications.
- ii) Write about Standard error of mean and its significance.
- iii) Discuss different types of Probability distribution.
- iv) Elaborate any Pharmaceutical example for Data analysis using SPSS.
- v) Discuss Null hypothesis, type I and type II Errors.
- vi) Explain Regression analysis.
- vii) Define and explain Experimental Study designs.
- viii) Explain Karl Pearsons Coefficient of correlation with examples.
- ix) Explain applications of Factorial designs in Pharmaceutical Product Development.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - January - 2023

Course: B. Pharm.

Semester: VIII

Subject: Biostatistics & Research Methodology

Subject Code: BP801T

Maximum Marks: 75

Duration: 3hrs

Date: 10.02.2023

Time: 2.00 PM to 05.00PM

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

1. Objective Type Questions: (Answer all the questions)

(10x2)=20

- i. What are descriptive statistics, what it includes?
- ii. Justify need of design for experiment.
- iii. Give any two absolute measures of dispersion.
- iv. Explain the correlation and causation effect.
- v. What is cohort's studies, for what it is used for?
- vi. What is regression analysis? differentiate simple and multiple regression.
- vii. What are exhaustive cases? give exhaustive cases for 3 coins tossing.
- viii. Define probability, 12 pillows in the bed, 6 are red, 4 are yellow and 2 are blue. What is the probability of picking a yellow pillow?
- ix. Differentiate between parametric and non-parametric tests.
- x. What is blocking and confounding effect in factorials?

1. Long Answers: (Answer 2 out of 3)

(2x10)=20

- a) What is frequency distribution? explain all measures of central tendency and measures of dispersion with example.
- b) What is correlation and causation effect? explain Karl Pearson's coefficient of correlation method.
- c) Group of 5 students studying in physics, chemistry, and biology stream, using random method of sample selection an intelligence test was administered to them & the scores obtained as under-

S. No.	Physics	Chemistry	Biology
1	15	12	12
2	14	14	15
3	11	10	14
4	12	13	10
5	10	11	10

Justify null hypothesis that students do not differ in their intelligence

2. Short Answers: (Answer 7 out of 9)

(7x5)=35

- a) What is probability, random event, sample space, mutually exclusive events, and independent events?
- b) What are distribution free tests, explain sign test and Wilcoxon signed rank test
- c) What is sampling, its objectives and explain the importance of sampling in hypothesis testing
- d) Elaborate on 't' test statistics for given data-

Subjects	1	2	3	4	5	6	7	8	9	10	11
Test I Marks	3	3	3	12	15	16	17	19	23	24	32
Test II Marks	20	13	13	20	29	32	23	20	25	15	30

- e) Distinguish between correlation and regression analysis and indicate the utility of regression analysis in pharmaceutical research.
- f) What are advantages of factorial designs, explain 2^3 factorial designs.
- g) Give importance of graphs in pharmaceutical research, explain Counter Plot graph
- h) Explain the importance of statistical software's in pharmacy and describe SPSS.
- i) Describe report writing procedures and presentation of data.

----- END OF PAPER -----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - January - 2023

Course: B. Pharm.

Subject: Biostatistics & Research Methodology

Subject Code: BP801T

Maximum Marks: 75

Date: 28/01/2023

Semester: VIII

Duration: 3hrs

Time: 2.00 PM to 05.00PM



Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Differentiate between Null and alternative hypothesis.
- ii) Enumerate Statistical Software's.
- iii) Enlist Advantages and disadvantages Pie charts.
- iv) Classify observational studies.
- v) Give two methods of sample size calculation in research study.
- vi) Write Applications of SAS or MINITAB.
- vii) Enlist non-parametric tests.
- viii) Define research and enlist its types.
- ix) Write significance of sampling in research.
- x) Define bias in clinical study.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What is hypothesis? What are different types of hypothesis? Explain how you will formulate a hypothesis with a suitable example.
- ii) Explain with suitable examples regression analysis and standard error of regression.
- iii) What are measures of dispersion? Explain their significance with suitable examples.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) What is Plagiarism? What are the needs for research?
- ii) Explain a typical experimental study design.
- iii) Write a short note on central composite design.
- iv) What do you mean by factorial design? Write some advantages of factorial design.
- v) Discuss null hypothesis, type I and type II errors.
- vi) Explain ANOVA and its applications.
- vii) Discuss the applications of EXCEL and SPSS programmes in statistical analysis
- viii) Describe the various graphical methods of representing quantitative data.
- ix) What are non-parametric tests? Explain chi square test-Goodness of fit test.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Winter 2023

Course : B. Pharmacy

Date : 02/01/2024

Subject Name : Biostatistics & Research Methodology

Sem : VIII

Max Marks : 75

Subject Code : BP801T

Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define Mean, Median and Mode.
- ii) Give need of research
- iii) Enumerate different graphs used for representing quantitative data
- iv) Enlist methods of sample size calculation in research study
- v) Give advantages and disadvantages of Pie chart.
- vi) Define bias in clinical studies.
- vii) Write formula for calculation of standard deviation and coefficient of correlation
- viii) Differentiate between type -I & type -II errors.
- ix) Enlist various online statistical software used in clinical trials
- x) Define a) Biostatistics b) Frequency distribution

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Discuss in detail Parametric tests.
- ii) Explain in detail about how statistical analysis is conducted by using Excel, SPSS & MINITAB.
- iii) Write a note on Response surface methodology.



Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Define hypothesis. Explain different types of hypothesis.
- ii) Explain Plagiarism? Give method to avoid Plagiarism.
- iii) Explain Multiple correlation with examples.
- iv) Explain in detail about Normal distribution, Poisson's distribution.
- v) Explain in short Mann - Whitney U test.
- vi) What do you mean by cohort studies? Write some advantages and disadvantages of cohort studies.
- vii) Describe design and analysis of experiments
- viii) Explain significance of measures of dispersion with suitable examples.
- ix) Write a short note on Factorial design.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: VIII

Subject Name: Biostatistics and Research Methodology

Subject Code: BP801T

Max Marks: 75

Date: 25-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Define Biostatistics & give its application in medical science.
- ii) Enlist different Parametric & Non parametric test used for hypothesis testing in research.
- iii) Differentiate between Central Composite Design (CCD) & Box-Behnken Design (BBD) in Design of Experiment.
- iv) Define Range & Standard Deviation.
- v) Write different properties of Normal Distribution Curve.
- vi) Explain Type I & Type II errors in Hypothesis testing.
- vii) Mention different values of correlation coefficient (r) to be interpreted as Very high, High, Moderate and Slight correlation.
- viii) What is Simple Random Sampling? Give examples.
- ix) Find the Median from data of tablet weights in mg as 110, 95, 102, 111, 99, 100.
- x) Give differences between Null Hypothesis and Alternative hypothesis

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Discuss about structure & content of research report. Describe different techniques used for presentation of data in research report.
- ii) What is Discrete Probability Distribution? Explain in brief about Binomial Distribution & Poisson Distribution.
- iii) Explain Karl Pearson Coefficient Correlation. Calculate the coefficient of correlation between sale of two competitive pharmaceutical products.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Product A (Lakh)	65	40	35	75	63	80	35	20	85	65	55	33
Product B (Lakh)	30	55	68	28	76	25	80	85	20	35	45	65

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) What is Historical Research Design? Describe its different steps.
- ii) How Clinical research can be classified? Explain with examples
- iii) Why research design is needed? Comment on various stages of research design.
- iv) Describe in detail about Non-parametric Kruskal-Wallis's test & Wilcoxon Rank Sum test.

- v) Determine Mode of following data by algebraic formula & represent it on graph.

Profit (Lakh)	5-10	10-15	15-20	20-25	25-30	30-35
No. of days	8	18	27	21	15	11

- vi) What is Plagiarism? Write about its different types.
- vii) Discuss the importance of SPSS, MINITAB and R-Software used for statistical analysis in research.
- viii) What is Ogive? Determine less than & more than Ogive of following data & represent graphically.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Students	4	7	9	10	15	22

- ix) Write in brief about Graphical presentation of data as Histogram, Pie chart and Response surface plot.

*** END OF THE PAPER ***